# Change the mass and watch the acceleration

**GUIDED NOTES / use with the projected diagram** Name _____

## A SAY THE WORD

### Fair test / ES: prueba controlada / PT: teste controlado

Change one thing at a time.

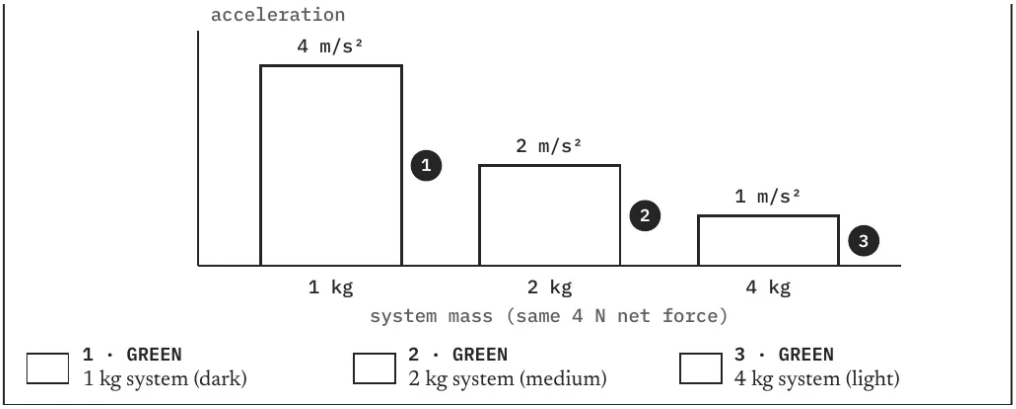
Also on your note page: Changed variable, Controls, Evidence.

## B READ THE RELATIONSHIP

### To test how mass matters, change mass and keep the net force the same.

At fixed net force, more mass means less acceleration

## C TRACE THE ORIGINAL FIGURE




Use the numbered key. Trace one feature. Tell your partner what it means.

## D FOLLOW THE WORKED EXAMPLE

### In one trial, velocity rises from 0 to 4 m/s in 2 s. Find the acceleration.

**Given** $v_i = 0$ m/s $v_f = 4$ m/s $\Delta t = 2$ s

**Rule** $a = \Delta v \div \Delta t$

**Work** $\Delta v = 4 - 0 = 4$ m/s $a = 4 \div 2$

**Answer** 2 m/s² use your own measured graph

## E SAVE FOR THE FINAL MEETING

A massive object is hard to nudge. The bigger XJ's mass, the smaller the acceleration any push or pull gives it, which matters for any plan to deflect it.

# Read the physics: Fair test

SUPPORTED READING / read one part, then point to the evidence Name _____

## 01 PICTURE IT

Compare a light cart with a loaded cart. Use the same net force and the same release method. Otherwise a change in acceleration could have more than one cause.

## 02 HOW IT WORKS

Read acceleration from the slope of a velocity–time graph. At fixed net force, doubling the moving mass halves acceleration. Multiplying mass by acceleration gives the same net force in both trials.

## 03 WHAT TO CHECK

Keep the hanging mass, track and release conditions consistent. Name what changed and what was controlled before stating the result.

## 04 TRY THE EXAMPLE FROM YOUR NOTES

In one trial, velocity rises from 0 to 4 m/s in 2 s. Find the acceleration. 2 m/s² use your own measured graph.

## 05 CONNECT TO XJ

A massive object is hard to nudge. The bigger XJ's mass, the smaller the acceleration any push or pull gives it, which matters for any plan to deflect it.

SAY IT BEFORE YOU WRITE

Which quantity must match in the two trials? We changed ___. We kept ___ fixed. Acceleration ___.

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Fair test

ES: prueba controlada / PT: teste controlado

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain fair test to a friend?<br>In my words, this means ___. | To test the effect of mass, what will you change and what will you keep the same?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How could you compare a light cart and a heavier cart fairly?<br>My example shows ___. | One run uses a heavier cart and a stronger pull. Why can't you tell what mass did?<br>I would change ___ because ___. |

## Changed variable

ES: variable que se cambia / PT: variável que se altera

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain changed variable to a friend?<br>In my words, this means ___. | When you add blocks to the cart to test the effect of mass, what did you change?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How could you label the runs so your partner can see exactly what varied?<br>My example shows ___. | The release point drifted accidentally. Is that the variable you meant to test?<br>I would change ___ because ___. |

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Controls

ES: variables controladas / PT: variáveis controladas

| Meaning in my words | How can you tell? |
|---|---|
| What are the controls in an experiment?<br>In my words, this means ___. | You test different cart masses. Why use the same hanging weight each time?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Name two things you would set the same before each release.<br>My example shows ___. | A partner lists cart mass as a control while adding mass each trial. Where does it belong?<br>I would change ___ because ___. |

## Evidence

ES: evidencia / PT: evidência

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain evidence to a friend?<br>In my words, this means ___. | What reading supports the claim that the cart reached 4 m in 2 s?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How could two measurements support a claim that one cart accelerated more?<br>My example shows ___. | “I expected our cart to win.” Does that tell us which cart crossed first?<br>I would change ___ because ___. |

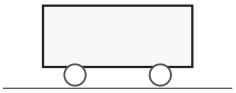
# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

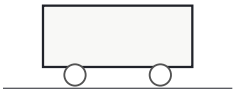
## Draw two masses with the same force. Label what changes, what stays fixed, and which accelerates more.

Labels you can use: change · control · compare.

Lower mass



Higher mass



**Tell a partner**

We changed ___. We kept ___ fixed. Acceleration ___.

# On assignment: Fair test

SECTION E / BACK / FINAL MEETING ONLY Name _____

**Two asteroid models get the same net force for the same time. Model H has twice the mass of L.**

## PICK A JOB / DELIVER ONE PRODUCT

☐ **Test engineer**

Build a two-row test card. Pick the model with the larger velocity change and show the mass comparison for the test crew.

☐ **Lab correspondent**

Explain to viewers why two equal pushes can look different. Your producer must check what stayed the same in both trials.

**Word bank: change · control · compare.**

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

## INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

**“We first said ___. Now we ___ because ___.”**